

BUS 445 | CUSTOMER ANALYTICS

Storytelling with Data

The Group Project Journey

SFU Beedie School of Business

Fall 2026

1. Project Overview: Your Role as a Strategic Consultant

The Storytelling with Data project is the capstone experience of BUS 445. It is a semester-long, simulated consulting engagement in which you apply the core principles of this course to a real-world business challenge, working as a team from the first framing of the problem through to a final, defensible recommendation.

Your role is not to be a data technician. You are a strategic advisor. Your client has a meaningful business problem but needs your team to frame the right analytics question, uncover the story in the data, and translate that story into a clear, persuasive, and actionable recommendation. Every model output you produce must ultimately translate into a dollar a senior decision-maker can act on.

Two Teams per Case, Two Sponsors, One Budget

Each case that runs in your section is assigned to exactly two teams, working independently on the same data. The two teams advise **different executives at the same client**. Each team has a named **primary sponsor**: the person who will sign your recommendation. A third executive, the **skeptic**, sits in on both engagements and will test both teams.

In the final session, the client can fund only one of the two recommendations next year. The two teams make the case for theirs in a moderated boardroom debate. The lesson is the heart of this course: the question determines the value of the answer.

2. How This Project Is Evaluated in Spirit

Strategic thinking and communication matter as much as technical execution. Three commitments shape everything you submit:

- **Strategy precedes statistics.** The most important work happens before any code is written. A precise, well-justified question is worth more than a sophisticated model aimed at the wrong target.
- **Justify your restraint.** You are rewarded for the tools and questions you deliberately set aside, not only for the breadth of what you attempt. A focused analysis that does one thing well beats a kitchen-sink approach every time.

- **End in a dollar.** Every recommendation closes with a **One Number** and an **Assumption Ledger** that shows where the number comes from and how fragile it is (Section 5).

3. The Project Journey

The project unfolds in a conceptual phase, a hands-on analysis phase, and a final synthesis. Deadlines are the same for both sections.

When	Phase	What Happens
Week 5	Kickoff and Assignment	Each team receives its case brief, its sponsor, and the dataset description. Read them together and meet as a team.
Weeks 5 to 7	Conceptual Phase	You do not yet have the raw data. Deconstruct your sponsor's decision, brainstorm hypotheses, and use the data dictionary to imagine what stories the data could tell. The Week 7 Model Bake-Off will inform which tools you keep and which you reject.
Sun Oct 25, 23:59	MILESTONE 1: Problem Definition Canvas (5%)	A one-page strategic plan submitted on Canvas before any modeling begins.
Mon Oct 26	Data Release	The full dataset is posted on Canvas the day after Milestone 1 closes, giving you five full weeks with the data.
Weeks 8 to 12	Discovery and Modeling	Explore the data, test your hypotheses, build your models, and build the dollar case. Week 10 is also the working window for the individual assignment (due Sun Nov 15), so plan team work around it.
Week 9	Canvas Feedback	Written feedback on your Canvas, including which of your two questions looks stronger. With the data in hand, confirm your question or switch to your backup.
Sun Nov 29, 23:59	MILESTONE 2: Memo and Recorded Presentation	Strategic memo and recorded boardroom presentation submitted on Canvas.
Week 13	MILESTONE 2: Comparative Debate Room	The two teams on each case defend competing recommendations for the client's single budget.

4. Milestone 1: The Problem Definition Canvas

5% of the course grade. Due Sunday, October 25 at 23:59 on Canvas. One page, produced before you have the raw data. Working only from the case brief and the dataset description, your team commits to a direction while keeping one alternative in reserve: a primary analytics question that boxes 4 to 9 are built around, and a backup you can switch to if the data does not support the first. The Canvas is graded on framing, not results; you do not need the data to do it well. It has nine boxes:

#	Box	What it must contain
1	Sponsor and decision	Who signs, and what will they do differently on Monday if they accept your recommendation?
2	Business objective	The dollar quantity the decision moves: revenue retained, partner fees earned, net operating income, cost avoided.
3	Analytics questions	Two candidate questions, ranked, each one sentence and labelled Oracle (who will do what) or Microscope (why, and what would change it). Both must serve your sponsor's decision. For the backup, add one line: what you would need to see in the data to switch to it.
4	Analytical pathway	For your primary question: the unit of analysis, the fields you will use, and the method you plan to apply.
5	Tools selected	The tools you will use and why each fits the question.
6	Tools rejected	At least two tools or questions you are deliberately setting aside, and why. This is the Anti-Kitchen-Sink Rule.
7	Success metrics	One business metric in dollars and one model metric, and how the model metric connects to the decision.
8	Assumption Ledger (draft)	The three assumptions your dollar figure will depend on most, and where each number will come from.
9	The skeptic's question	The hardest question your case's skeptic will ask, and how you plan to answer it.

Any visual layout is fine as long as all nine boxes fit on one page. Written feedback is returned in Week 9.

5. Milestone 2: Memo, Recorded Presentation, and Debate

20% of the course grade. Three connected deliverables.

The Strategic Memo (due Sunday, November 29 at 23:59)

A polished consulting report of **10 to 12 pages**, excluding appendices, written for your sponsor. It should be a document you would be proud to show a future employer. It must include:

- **Page 1 executive summary** that states the recommendation and the One Number before anything else.
- The sponsor's decision and your final analytics question, why it is the one worth answering, and, if you switched from your Canvas primary, what in the data made you switch.
- Data and method, including the tools you rejected and why.
- Findings, told as a story rather than as a list of model outputs.

- The dollar case: the One Number, the Assumption Ledger, and how the One Number moves if your most important assumption is wrong.
- An implementation and validation plan: how the client will know, after the fact, that your recommendation worked. Where your evidence is predictive or stated rather than causal, propose the test that would settle it.
- Appendices: your AI use note (tools used and the prompts that guided your work), plus any supporting tables. Submit your R script as a separate file.

The One Number and the Assumption Ledger

The One Number is the net dollar impact your recommendation creates for the client over the next 12 months, after the cost of carrying it out. Both teams on a case use this same definition, so the client can compare them directly.

The Assumption Ledger is a short table listing every input to the One Number that does not come from the data (margins, costs, response rates, base rates, occupancy), the value you used, its source or justification, and how much the One Number changes if that input is off by half. The debate will go straight to this table.

The Recorded Boardroom Presentation (due Sunday, November 29 at 23:59)

An **8 to 10 minute** recorded presentation to your sponsor and the skeptic: senior, intelligent, non-technical. Persuasion, clarity, and impact matter more than technical exposition. **Every team member speaks.** Recordings are reviewed by the instructor only and are released to the class after both sections have debated.

The Comparative Debate Room (Week 13)

The motion for every case is the same: **the client can fund one of the two recommendations next year. Which one should it be?** Because your full case is already on video, class time goes to cross-examination, not exposition. Each case gets the same format in both sections:

Segment	Time	What happens
Opening position	2 min per team	Your recommendation and your One Number. No slides.
The tensions	About 20 min	The instructor, sitting as the skeptic, puts the sharpest tensions between the two memos on the table. Each team also puts one prepared question directly to the other team.
The floor	About 10 min	The rest of the class, acting as client leadership, challenges both teams.
Closing	1 min per team	Restate your One Number and why the client should fund your recommendation.

You will receive the other team's memo **two days before your debate**, so your prepared question can go to the heart of their argument. Any member of your team may be asked to answer; prepare as a

team, not as one speaker with support. When you are in the audience, the quality of your challenges to other teams counts toward your participation grade.

6. How Your Project Will Be Graded

Component	Weight	Due
Milestone 1: Problem Definition Canvas	5%	Sun Oct 25, 23:59
Milestone 2: Strategic Memo	10%	Sun Nov 29, 23:59
Milestone 2: Recorded Presentation	4%	Sun Nov 29, 23:59
Milestone 2: Comparative Debate (team)	6%	Week 13
Total	25%	

Both milestones are evaluated holistically against the criteria below.

Criterion	What We Are Looking For
Question Framing and Justification	How well you framed your sponsor's decision as a valuable analytics question and justified why it is worth answering.
Appropriateness and Rigor of Method	Whether you selected and executed the right methods, and whether you can explain the ones you rejected.
Depth of Data Exploration	How insightful and thorough your investigation of the data was.
Business Value and the "So What?"	How well you translated findings into a clear One Number, with an honest Assumption Ledger and sensitivity.
Storytelling and Persuasion	How compelling and well structured your narrative was in the memo, the video, and the debate.
Defending Under Challenge	Your ability to defend your choices against the skeptic, the other team, and the floor.
Clarity and Professionalism	The quality of your writing, visualizations, and overall presentation.
Collaboration and Teamwork	Assessed through a confidential peer evaluation, due Sun Dec 6 at 23:59. Where contributions were clearly unequal, it is used to adjust individual Milestone 2 grades.

7. Ground Rules

- **Independence.** The two teams on a case work independently. Do not share code, findings, or drafts with the other team on your case, or with any team in the other section.

- **Debate confidentiality.** Do not discuss your debate, or the questions raised in it, with students in the other section until both sections have debated. This is covered by the course academic integrity policy.
- **AI as co-pilot.** Use AI tools freely as coding accelerators and brainstorming partners. Disclose how you used them, verify everything they produce, and keep the analytical judgment your own.
- **Data.** Work from the data released on Canvas. If your case allows optional external data, record exactly which version you used and when you downloaded it.

A final word. Treat this project as the engagement you would want on your résumé. The data is real, the questions are open, and the most valuable skill you will practice is not running a model. It is asking a better question than the team next to you, and convincing the room that the client should fund yours.